Damla Görgülü

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Education

University of Cambridge, Cambridge, United Kingdom 10/2026 – Present
MPhil in Electrical and Electronic Engineering

- **Michaelmas modules:** Photonic Devices; Very Large Scale Integration (VLSI); Electric Vehicles and Transport Electrification; Integrated Electronic Devices.
- **Further planned modules:** Quantum Computing, Communications and Emerging Technologies; Entrepreneurship, University Spin-offs and IP Management; Sensor, Embedded and Control Systems; Bioelectronics.
- **Scholarship:** Turkish Education Foundation (UK) Cambridge Masters Scholarship (Cambridge Trust).

Koç University, İstanbul, Turkey
Bachelor of Computer Engineering (Double Major), GPA 3.87/4.00 09/2022 – 02/2026
Bachelor of Electrical and Electronics Engineering, GPA 3.87/4.00 09/2021 – 07/2025

- **Specialisation:** Artificial Intelligence; graduated summa cum laude.
- **Awards:** Dean's Honour Roll; Vehbi Koç Honour Award (eight consecutive semesters).

Publications and Manuscripts

- **D. Görgülü**, F. A. Güler, A. B. Trabzon and M. C. Onbaşlı, "Benchmarking Skyrmion Logic Against Foundry CMOS Circuits", manuscript submitted to Nanoscale, 2026. The first three authors contributed equally.

Research Experience

Researcher, Koç University — Hypernetwork-Based LoRA Generation 06/2025 – Present

- Research **hypernetwork-based LoRA generation** in a four-person team led by Assoc. Prof. Aykut Erdem, developing adapters for content and style fusion in diffusion models.
- Contribute to model training and experimental evaluation of generated LoRA adapters for image synthesis.

Researcher, Koç University — Spintronics and CMOS 02/2025 – 09/2026

- Co-developed an **RTL-to-skyrmion benchmarking workflow** for a 4-bit ALU, 8-point DFT engine and multilayer perceptron, comparing SKY130 CMOS implementations with synthetic-antiferromagnetic skyrmion logic.
- Quantified area, latency and energy trade-offs; mapped skyrmion networks showed **3.44–4.06× smaller core-logic footprints** than CMOS, excluding peripheral circuitry.
- Communicated the benchmark methodology and findings through a co-authored research manuscript.

Researcher, Technical University of Munich, Chair of AI Processor Design 03/2026 – 06/2026

- Designed **weight and bias partitioning** and transfer schedules to improve memory-bandwidth utilisation in neural-network accelerators.
- Implemented **S-Link streaming transfers** between a Cheshire RISC-V SoC and a ButterStick FPGA to support neural-network inference using external DRAM.

Researcher, Technical University of Munich, Chair of AI Processor Design 07/2025 – 12/2025

- Brought up **LiteX-based RISC-V SoCs** on OrangeCrab and ButterStick FPGAs and established a Linux-booted SD-card-to-DRAM path for hardware experiments.
- Integrated custom RTL counters and memory-processing modules using control and status registers (CSRs) to verify SoC memory and I/O interfaces.
- Investigated DRAM retention and refresh behaviour and presented experimental findings in technical reviews.

Researcher, Technical University of Munich, Chair of AI Processor Design 06/2024 – 09/2024

- Investigated **BTI/HCI transistor ageing** under Prof. Hussam Amrouch using the CARAT framework, examining reliability effects on AI processor circuits.
- Adapted the framework to **3 nm and 14 nm** technology nodes, simulated Verilog adders, multipliers and multiply–accumulate units, and analysed ageing-induced delay variation using MATLAB and Synopsys WaveView.

Undergraduate Researcher, Koç University — RISC-V Microprocessor Design 06/2023 – 09/2024

- Co-developed a **RISC-V-based CNN inference processor** for MNIST classification with two peers, implementing digital microprocessor and accelerator modules in **VHDL and Verilog**.
- Quantised CNN weights to Q0.15, integrated UART communications and developed a Python interface for input testing and verification.
- Selected as a finalist in the **TEKNOFEST 2024 Microprocessor Design Competition**.

Computer Vision Intern, Savunma Teknolojileri ve Mühendislik A.Ş. (STM) 07/2023 – 08/2023

- Researched **GAN-based licence-plate super-resolution** and trained Real-ESRGAN on **3,500+ images**, comparing paired and dynamically generated datasets.
- Evaluated **4× and 16×** super-resolution outputs on drone imagery from field tests.

Selected Teaching Experience

Tutor, Koç University KOLT Tutoring Center 02/2024 – 06/2024

- Supported **120+ Digital System Design students** through weekly office hours; delivered three weekly office hours and review sessions in Electromagnetism, explaining engineering concepts through worked problems.

Teaching Assistant, Koç University — Signals and Systems 02/2023 – 06/2023

- Supported **50+ students** in a four-assistant team, leading weekly problem-solving sessions and office hours and explaining mathematical derivations in small-group discussions.

Research Communication and Leadership

Co-Founder and Editor, Koç University Undergraduate Journal of Engineering 09/2023 – 09/2024

- Co-founded an editorial initiative to showcase undergraduate engineering research and contacted students undertaking theses to invite submissions.
- Edited student research writing and coordinated with contributors to make technical work accessible to a wider engineering audience.

Skills

- **Hardware and EDA:** VHDL, Verilog, FPGA/SoC design, RISC-V, LiteX, OpenLane/SKY130, Synopsys WaveView.
- **Programming and Tools:** Python, C/C++, Java, MATLAB, Linux/Unix, Git.
- **Languages:** Turkish (Native), English (C2), Spanish (A2–B1), German (A1).